

Social learning by type-token frequency - analysis

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Abstract

Abstract (to be written). Argument structure - See below. Analysis structure - 1. Descriptives of difference scores and tests against zero. 2. GLMMs, with reference to ANOVA. Discuss quorum sensing [Sumpter and Pratt, 2009] Discuss moral repetition effect, and tolerance principle and submit to psych sci. In intro. Different formal perspectives suggest that mere frequency should not be relevant, From a learning perspective, [Explanation of tolerance principle; argue later that threshold is $8/\ln(8) = 3.8$, but that we did not vary this experimentally as it's too long; general explanation of tolerance principle- When having a productive rule is faster than listing all items as exception]. From a representational perspective, dispositions. intrinsic properties of banana; bayesian argument. From a social perspective, majority. Whatever the moral frequency guy argues.

1 Methods (continued)

1.1 Exclusions

I excluded 0 participants because they failed the attention test.

1.2 SD3 REMOVE THIS IN PAPER

We next analyze the SD3 and combine the results with the data frame for the main experiment.

Table 1: Number of participants in each group of the SD3 scales. Note that this applies to city students only

scale	N			Percentage		
	high	low	NA_	high	low	NA_
Machiavellianism	23	24	5	0.442	0.462	0.096
Narcissism	26	26	0	0.500	0.500	0.000
Psychopathy	25	24	3	0.481	0.462	0.058

Table 2: Demographics of the final sample

population	N	Females	Males	Other	Age	
					Age.m	Age.range
students	52	48	3	1	18.5	18-22
testable	119	59	59	1	32.7	18-61

1.3 Analysis strategy

I analyzed the data in three stages. First, I present descriptives of the raw ratings to serve as task validations, as good actions should be rated as morally better than bad actions. However, I had no other predictions for the raw levels of acceptability.

Second, for each domain (first-party acceptability, third-party acceptability, prevalence), I calculated two types of difference scores. Following [Rosenthal et al., 1999], I used focused difference scores (rather than models with effects for which I had no predictions) to target the contrasts of interest, also ensuring consistency between visualizations and statistical tests. The first type of difference score assesses the effect of type and token frequency compared to single occurrence events, namely $\frac{\text{High Type-Frequency}-\text{Single Occurrence}}{\text{High Type-Frequency}+\text{Single Occurrence}}$ and $\frac{\text{Low Type-Frequency}-\text{Single Occurrence}}{\text{Low Type-Frequency}+\text{Single Occurrence}}$. I compared these difference scores to the chance level of zero using Wilcoxon tests.

The second type of difference score compared the effects of type and token frequency with each other, specifically $\frac{\text{High Type-Frequency}-\text{Low Type-Frequency}}{\text{High Type-Frequency}+\text{Low Type-Frequency}}$. Again, I compared these difference scores to the chance level of zero. For both types of difference scores, I also asked whether they would be affected by the valence. P-values were corrected using the Holm-Bonferroni method.

Finally, I used a series of Cumulative Linear Mixed Model (CLMM) with a logit link function to confirm these results [Bürkner and Vuorre, 2019, Liddell and Kruschke, 2018], using the ordinal *R* package [Christensen, 2023]. CLMMs assume that participant responses reflect an underlying latent variable. The model estimates multiple thresholds corresponding to different response categories, with a response given when the latent variable exceeds the threshold for that category. In addition to the thresholds, they also estimate linear predictors influencing the latent variable.

NOTE TO SELF: THERE ARE ALSO SEQUENTIAL MODELS, SEE [Bürkner and Vuorre, 2019].

Problems with treating ordinal responses [Liddell and Kruschke, 2018]

<https://pmc.ncbi.nlm.nih.gov/articles/PMC10439063/> [Taylor et al., 2023] this is about norms

1.4 Participants

2 Results

2.1 Raw ratings

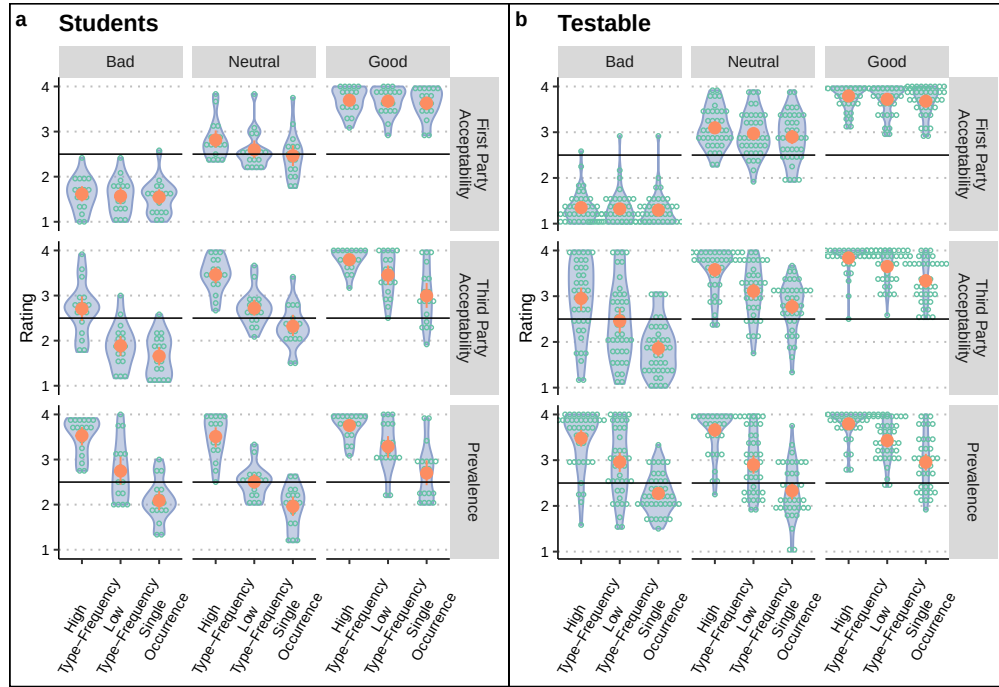


Figure 1: Ratings for (a) testable participants and (b) students. The dots, error bars and violin represent the sample average, 95% bootstrap confidence intervals and the distribution of the individual responses, respectively. Open circles represent individual participants.

The raw ratings are shown in Figure 1 and Table 5. As shown in Table 6, a CLMM with a logit link function, including the fixed predictors Valence and Frequency Condition, their interaction, and a random slope for participants—showed that, for both acceptability questions and both samples, good deeds were judged more positively than neutral ones, and neutral ones more positively than bad ones. The valence manipulation was thus successful.

2.2 Comparisons of the high and low type-frequency conditions against the single occurrence condition

I next turned to the question of whether type-frequency or token-frequency would affect acceptability and prevalence judgments, using the difference scores that compared each frequency condition to the single-occurrence condition.

2.2.1 Difference scores

The results are shown in Figures 2 and Table 7. For first-party acceptability, ratings in the high type frequency condition differed reliably from those in the single occurrence condition, mainly for good and neutral actions. For bad actions, while the uncorrected effect was significant, it became marginal after correction in the testable sample, and non-significant in the student sample. In contrast, ratings in the low type frequency condition did not significantly differ from those in the single occurrence condition. First-party acceptability thus seems primarily driven by type rather than token frequency.

To provide evidence for the null hypothesis, I fit two linear models to the first-party acceptability difference scores: one representing the null hypothesis (with the mean fixed at 0) and one representing the alternative hypothesis (with the mean estimated from the data). I then calculated the likelihood of the data under each model, applied corrections for the numbers of parameters using the Bayesian Information Criterion (BIC) and the Akaike Information Criterion (AIC), and computed the likelihood ratio, with values greater than one indicating support for the null hypothesis. As shown in Table 7, the first-party acceptability difference scores comparing the low type-frequency condition to the single-occurrence condition generally provided weak to moderate evidence in favor of the null hypothesis, suggesting again that first-party acceptability is primarily driven by type frequency rather than token frequency.

For both third-party acceptability and prevalence, ratings in both the high and low type frequency conditions differed from the single occurrence condition across all valence conditions, though the difference scores were roughly twice as large for the high type frequency condition compared to the low type frequency condition. Both type and token frequency thus affect these ratings, though I will demonstrate below that the effect of type frequency is substantially larger.

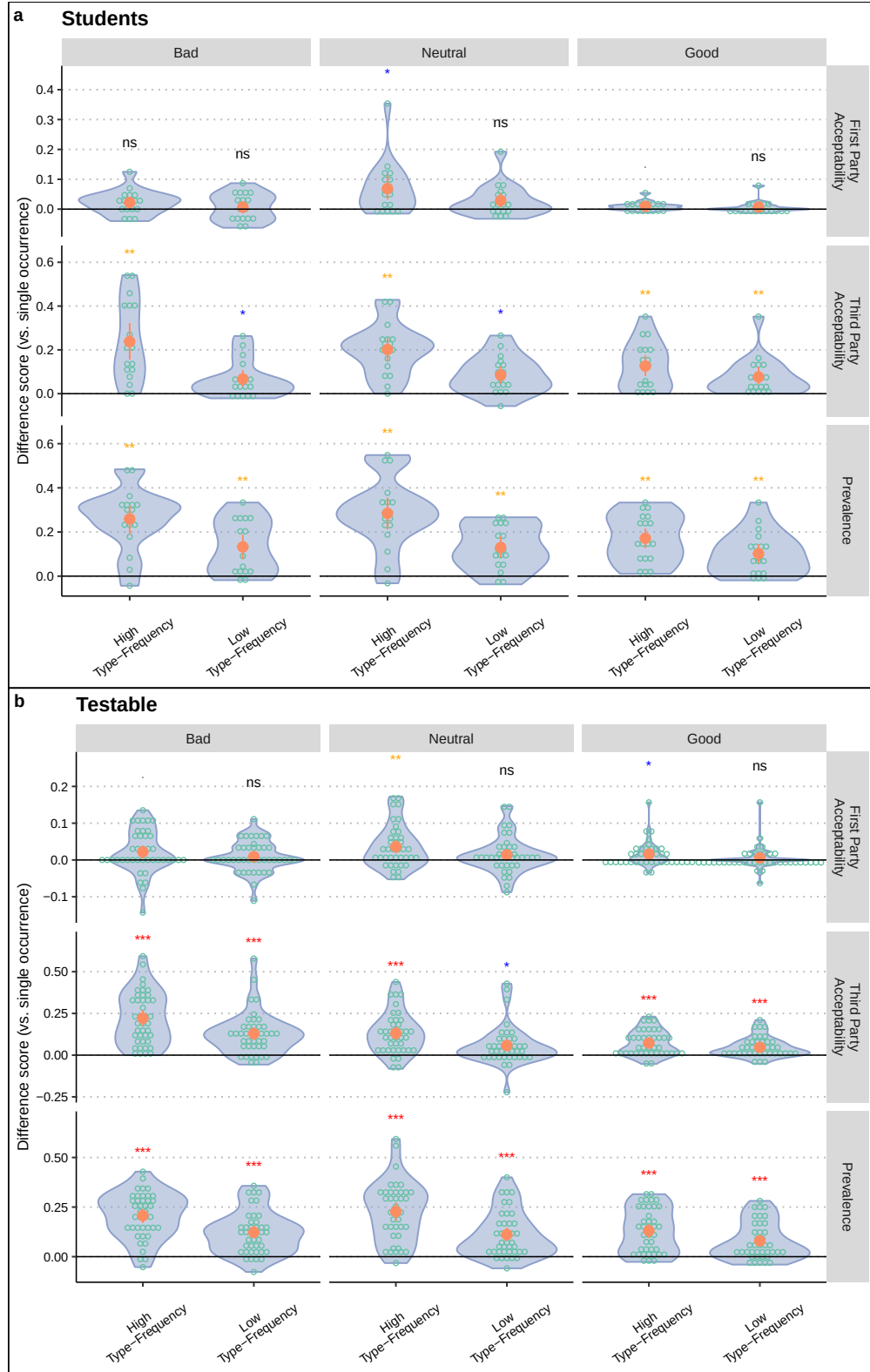


Figure 2: Difference scores against the single occurrence condition ($\frac{\text{High Type-Frequency}-\text{Single Occurrence}}{\text{High Type-Frequency}+\text{Single Occurrence}}$ and $\frac{\text{Low Type-Frequency}-\text{Single Occurrence}}{\text{Low Type-Frequency}+\text{Single Occurrence}}$). The dot, error bars and violin represent the sample average, 95% bootstrap confidence intervals and the distribution of the individual responses, respectively. Open circles represent individual participants. Significance codes: *** $p \leq .001$, ** $p \leq .01$, * $p \leq .05$, . $p \leq .10$, ns = not significant.

2.2.2 Confirmation with ordinal model

I confirmed these results with a series of CLMMs. For each domain (acceptability and prevalence), I included the fixed factor predictors Frequency Condition (High Type-Frequency vs. Single Occurrence and Low Type-Frequency vs. Single Occurrence) and Valence (with the reference level set to neutral), as well as a random slope for participants.

Table 3: Results from a set of cumulative link mixed models, fitted separately for each sample (testable and students), domain (First-Party Acceptability, Third-Party Acceptability, and Prevalence), and frequency contrast (High Type-Frequency vs. Single Occurrence and Low Type-Frequency vs. Single Occurrence), with fixed factor predictors Frequency Condition and Valence, their interaction, and a random slope for participants. Significance codes: *** $p \leq .001$, ** $p \leq .01$, * $p \leq .05$, . $p \leq .10$, ns = not significant.

Parameter	Coefficient	SE	CI_{low}	CI_{high}	z	p
First Party Acceptability - High Type-Frequency (vs. Single Occurrence) - Students						
Valence:bad	-2.024	0.113	-2.245	-1.803	-17.984	0.000 ***
Valence:good	2.891	0.113	2.670	3.112	25.614	0.000 ***
Freq. Cond:high Type-Freq.	0.757	0.087	0.587	0.927	8.726	0.000 ***
Valence:bad×Freq. Cond:high Type-Freq.	-0.602	0.105	-0.809	-0.396	-5.720	0.000 ***
Valence:good×Freq. Cond:high Type-Freq.	-0.516	0.109	-0.729	-0.303	-4.747	0.000 ***
First Party Acceptability - High Type-Frequency (vs. Single Occurrence) - Testable						
Valence:bad	-3.946	0.307	-4.548	-3.344	-12.852	0.000 ***
Valence:good	2.147	0.315	1.529	2.766	6.808	0.000 ***
Freq. Cond:high Type-Freq.	0.471	0.130	0.216	0.726	3.627	0.000 ***
Valence:bad×Freq. Cond:high Type-Freq.	-0.241	0.205	-0.643	0.161	-1.176	0.240 ns
Valence:good×Freq. Cond:high Type-Freq.	0.136	0.215	-0.286	0.559	0.633	0.526 ns
First Party Acceptability - Low Type-Frequency (vs. Single Occurrence) - Students						
Valence:bad	-2.094	0.435	-2.946	-1.242	-4.818	0.000 ***
Valence:good	2.940	0.443	2.072	3.807	6.642	0.000 ***
Freq. Cond:low Type-Freq.	0.303	0.193	-0.074	0.681	1.575	0.115 ns
Valence:bad×Freq. Cond:low Type-Freq.	-0.267	0.277	-0.810	0.276	-0.962	0.336 ns
Valence:good×Freq. Cond:low Type-Freq.	-0.149	0.300	-0.737	0.439	-0.497	0.619 ns
First Party Acceptability - Low Type-Frequency (vs. Single Occurrence) - Testable						
Valence:bad	-4.025	0.335	-4.681	-3.369	-12.022	0.000 ***
Valence:good	2.189	0.331	1.541	2.837	6.622	0.000 ***
Freq. Cond:low Type-Freq.	0.187	0.129	-0.065	0.440	1.453	0.146 ns
Valence:bad×Freq. Cond:low Type-Freq.	-0.082	0.208	-0.490	0.325	-0.397	0.691 ns
Valence:good×Freq. Cond:low Type-Freq.	0.002	0.207	-0.405	0.408	0.008	0.993 ns
Third Party Acceptability - High Type-Frequency (vs. Single Occurrence) - Students						
Valence:bad	-1.609	0.398	-2.389	-0.828	-4.040	0.000 ***
Valence:good	1.745	0.393	0.974	2.516	4.434	0.000 ***
Freq. Cond:high Type-Freq.	3.019	0.221	2.586	3.452	13.669	0.000 ***
Valence:bad×Freq. Cond:high Type-Freq.	-0.456	0.282	-1.009	0.097	-1.616	0.106 ns
Valence:good×Freq. Cond:high Type-Freq.	-0.225	0.310	-0.833	0.382	-0.727	0.468 ns
Third Party Acceptability - High Type-Frequency (vs. Single Occurrence) - Testable						
Valence:bad	-2.230	0.326	-2.868	-1.591	-6.841	0.000 ***
Valence:good	1.502	0.330	0.855	2.148	4.553	0.000 ***
Freq. Cond:high Type-Freq.	2.438	0.150	2.144	2.732	16.256	0.000 ***
Valence:bad×Freq. Cond:high Type-Freq.	0.282	0.194	-0.098	0.661	1.453	0.146 ns

Table 3: Results from a set of cumulative link mixed models, fitted separately for each sample (testable and students), domain (First-Party Acceptability, Third-Party Acceptability, and Prevalence), and frequency contrast (High Type-Frequency vs. Single Occurrence and Low Type-Frequency vs. Single Occurrence), with fixed factor predictors Frequency Condition and Valence, their interaction, and a random slope for participants. Significance codes: *** $p \leq .001$, ** $p \leq .01$, * $p \leq .05$, . $p \leq .10$, ns = not significant. (continued)

Parameter	Coefficient	SE	CI _{low}	CI _{high}	z	p	
Valence:good×Freq. Cond:high Type-Freq.	-0.182	0.227	-0.627	0.263	-0.800	0.423	ns
Third Party Acceptability - Low Type-Frequency (vs. Single Occurrence) - Students							
Valence:bad	-2.029	0.540	-3.087	-0.971	-3.760	0.000	***
Valence:good	1.966	0.532	0.922	3.009	3.692	0.000	***
Freq. Cond:low Type-Freq.	1.112	0.199	0.723	1.501	5.602	0.000	***
Valence:bad×Freq. Cond:low Type-Freq.	-0.392	0.281	-0.943	0.159	-1.395	0.163	ns
Valence:good×Freq. Cond:low Type-Freq.	0.333	0.283	-0.222	0.888	1.175	0.240	ns
Third Party Acceptability - Low Type-Frequency (vs. Single Occurrence) - Testable							
Valence:bad	-2.417	0.274	-2.953	-1.880	-8.833	0.000	***
Valence:good	1.557	0.287	0.994	2.120	5.421	0.000	***
Freq. Cond:low Type-Freq.	0.893	0.128	0.643	1.143	7.006	0.000	***
Valence:bad×Freq. Cond:low Type-Freq.	0.669	0.179	0.319	1.020	3.742	0.000	***
Valence:good×Freq. Cond:low Type-Freq.	0.247	0.193	-0.131	0.626	1.280	0.200	ns
Prevalence - High Type-Frequency (vs. Single Occurrence) - Students							
Valence:bad	0.493	0.361	-0.214	1.200	1.366	0.172	ns
Valence:good	2.125	0.362	1.416	2.834	5.872	0.000	***
Freq. Cond:high Type-Freq.	4.409	0.249	3.922	4.896	17.736	0.000	***
Valence:bad×Freq. Cond:high Type-Freq.	-0.607	0.290	-1.175	-0.039	-2.095	0.036	*
Valence:good×Freq. Cond:high Type-Freq.	-1.085	0.311	-1.695	-0.476	-3.491	0.000	***
Prevalence - High Type-Frequency (vs. Single Occurrence) - Testable							
Valence:bad	-0.155	0.276	-0.696	0.385	-0.564	0.573	ns
Valence:good	1.790	0.253	1.295	2.286	7.081	0.000	***
Freq. Cond:high Type-Freq.	4.255	0.162	3.938	4.572	26.317	0.000	***
Valence:bad×Freq. Cond:high Type-Freq.	-0.750	0.198	-1.138	-0.362	-3.789	0.000	***
Valence:good×Freq. Cond:high Type-Freq.	-1.112	0.206	-1.515	-0.708	-5.395	0.000	***
Prevalence - Low Type-Frequency (vs. Single Occurrence) - Students							
Valence:bad	0.541	0.522	-0.482	1.565	1.037	0.300	ns
Valence:good	2.519	0.521	1.499	3.540	4.839	0.000	***
Freq. Cond:low Type-Freq.	1.922	0.219	1.493	2.350	8.783	0.000	***
Valence:bad×Freq. Cond:low Type-Freq.	0.068	0.287	-0.495	0.631	0.236	0.813	ns
Valence:good×Freq. Cond:low Type-Freq.	-0.094	0.294	-0.670	0.482	-0.321	0.748	ns
Prevalence - Low Type-Frequency (vs. Single Occurrence) - Testable							
Valence:bad	-0.212	0.401	-0.998	0.573	-0.530	0.596	ns
Valence:good	2.043	0.409	1.241	2.845	4.993	0.000	***
Freq. Cond:low Type-Freq.	1.759	0.137	1.490	2.028	12.810	0.000	***
Valence:bad×Freq. Cond:low Type-Freq.	0.365	0.185	0.002	0.728	1.968	0.049	*
Valence:good×Freq. Cond:low Type-Freq.	-0.253	0.191	-0.627	0.121	-1.324	0.186	ns

The results are shown in Table 3. For first-party acceptability, ratings differed significantly from the single-occurrence condition only in the high type-frequency condition, confirming that first-party acceptability is

primarily driven by type frequency. (In the student sample, some additional interactions emerged, but these could not be replicated in the testable sample.)

For third-party acceptability and prevalence, both the high and low type-frequency conditions differed from the single-occurrence condition (except for the low type-frequency condition in third-party acceptability ratings within the student sample). However, the parameter estimates were approximately two to three times greater for the high type-frequency condition than for the low type-frequency condition.

For prevalence, the effect of high type-frequency was reduced for good and bad actions relative to neutral actions, while the effect of token frequency was slightly stronger for bad actions in the testable sample.

2.3 Comparisons of the high and low type-frequency conditions

I now turn to the question of whether type and token frequency affect ratings differentially. I first address this question using difference scores, and then confirm these results using an CLMM.

2.3.1 Difference scores

```
## Bin width defaults to 1/30 of the range of the data. Pick better value with
## `binwidth`.
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## `binwidth`.
```

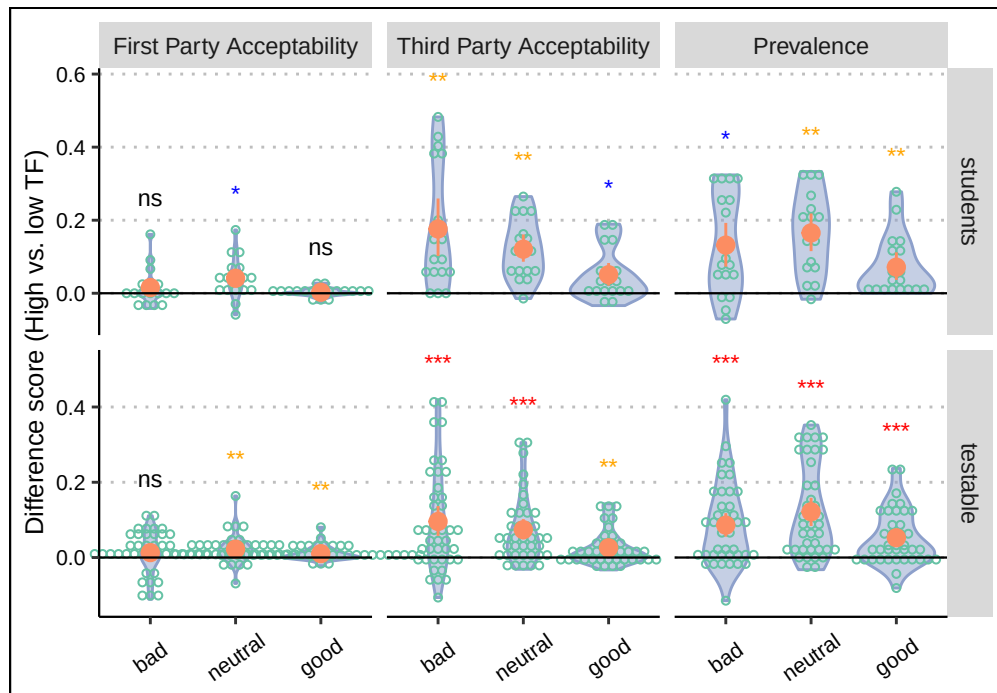


Figure 3: Difference scores comparing the High and Low Type-Frequency Conditions ($\frac{\text{High Type-Frequency} - \text{Low Type-Frequency}}{\text{High Type-Frequency} + \text{Low Type-Frequency}}$). The dots, error bars and violin represent the sample average, 95% bootstrap confidence intervals and the distribution of the individual responses, respectively. Open circles represent individual participants. Significance codes: *** $p \leq .001$, ** $p \leq .01$, * $p \leq .05$, . $p \leq .10$, ns = not significant.

The results are shown in Figure 3 and Table ???. For first-party acceptability, the difference scores were significantly greater than zero for good and neutral actions in the testable sample, and for neutral actions only in the student sample. First-party acceptability ratings thus increase when the type frequency of an action is increased, even when the frequency of occurrence is held constant.

For third-party acceptability and prevalence, the difference scores were significantly greater than zero for all valence conditions and samples, showing again that type-frequency has an effect over and above frequency of occurrence.

2.3.2 Confirmation with ordinal model

I confirmed these results with a series of CLMMs. For each domain, I fitted a model including the fixed factor predictors Frequency Condition (High Type-Frequency vs. Low Type-Frequency) and Valence, with the reference level set to neutral.

Table 4: Results from a set of cumulative link mixed models, contrasting the High Type-Frequency condition with the Low Type-Frequency condition. Models were fitted separately for each sample (online or student) and domain (Prevalence, First-Party Acceptability, and Third-Party Acceptability), with fixed factor predictors Valence, Frequency Condition, and their interaction, as well as a random slope for participants. Significance codes: *** $p \leq .001$, ** $p \leq .01$, * $p \leq .05$, . $p \leq .10$, ns = not significant.

Parameter	Coefficient	SE	CI_{low}	CI_{high}	z	p
First Party Acceptability - Students						
Valence:bad	-2.348	0.424	-3.179	-1.518	-5.544	0.000 ***
Valence:good	2.798	0.432	1.951	3.645	6.478	0.000 ***
Freq. Cond:high Type-Freq.	0.485	0.192	0.109	0.862	2.525	0.012 *
Valence:bad×Freq. Cond:high Type-Freq.	-0.364	0.277	-0.906	0.178	-1.315	0.188 ns
Valence:good×Freq. Cond:high Type-Freq.	-0.397	0.304	-0.993	0.199	-1.306	0.191 ns
First Party Acceptability - Testable						
Valence:bad	-4.139	0.327	-4.780	-3.498	-12.657	0.000 ***
Valence:good	2.172	0.330	1.525	2.819	6.577	0.000 ***
Freq. Cond:high Type-Freq.	0.272	0.131	0.016	0.528	2.079	0.038 *
Valence:bad×Freq. Cond:high Type-Freq.	-0.138	0.205	-0.540	0.264	-0.674	0.500 ns
Valence:good×Freq. Cond:high Type-Freq.	0.156	0.220	-0.276	0.588	0.709	0.479 ns
Third Party Acceptability - Students						
Valence:bad	-2.010	0.415	-2.824	-1.195	-4.837	0.000 ***
Valence:good	2.158	0.419	1.336	2.980	5.147	0.000 ***
Freq. Cond:high Type-Freq.	2.111	0.211	1.698	2.524	10.018	0.000 ***
Valence:bad×Freq. Cond:high Type-Freq.	-0.141	0.281	-0.691	0.410	-0.501	0.616 ns
Valence:good×Freq. Cond:high Type-Freq.	-0.549	0.316	-1.169	0.071	-1.736	0.083 .
Third Party Acceptability - Testable						
Valence:bad	-1.598	0.424	-2.428	-0.768	-3.772	0.000 ***
Valence:good	1.817	0.439	0.957	2.678	4.140	0.000 ***
Freq. Cond:high Type-Freq.	1.673	0.153	1.372	1.973	10.909	0.000 ***
Valence:bad×Freq. Cond:high Type-Freq.	-0.346	0.198	-0.734	0.042	-1.747	0.081 .
Valence:good×Freq. Cond:high Type-Freq.	-0.415	0.240	-0.886	0.056	-1.726	0.084 .
Prevalence - Students						
Valence:bad	0.699	0.440	-0.164	1.562	1.587	0.113 ns
Valence:good	2.336	0.445	1.464	3.208	5.252	0.000 ***
Freq. Cond:high Type-Freq.	2.932	0.222	2.496	3.368	13.189	0.000 ***
Valence:bad×Freq. Cond:high Type-Freq.	-0.656	0.291	-1.226	-0.086	-2.255	0.024 *
Valence:good×Freq. Cond:high Type-Freq.	-1.090	0.313	-1.704	-0.476	-3.479	0.001 ***
Prevalence - Testable						
Valence:bad	0.261	0.448	-0.617	1.140	0.583	0.560 ns
Valence:good	1.641	0.458	0.742	2.539	3.578	0.000 ***
Freq. Cond:high Type-Freq.	2.878	0.167	2.551	3.206	17.220	0.000 ***
Valence:bad×Freq. Cond:high Type-Freq.	-0.999	0.215	-1.419	-0.578	-4.655	0.000 ***
Valence:good×Freq. Cond:high Type-Freq.	-1.066	0.233	-1.522	-0.610	-4.583	0.000 ***

The results are shown in Table 4. In all domains and samples, ratings in the high type-frequency condition were higher than those in the low type-frequency condition.

We also observed main effects of valence, showing that good actions were rated higher than neutral ones, while neutral ones were rated higher than bad ones (except for prevalence ratings, where the latter two conditions did not differ). These results thus mirror those of the difference scores above.”

2.4 Alternative analyses

In SI 5, I report two alternative sets of analyses in addition to the difference score analyses and the CLMMs. First, I fit a series of generalized linear mixed models (GLMMs) using binarized responses, where ratings of 1 or 2 were coded as “unacceptable” and ratings of 3 or 4 as “acceptable.” These models compared both the High Type-Frequency (TF) and Low TF conditions to the Single Occurrence condition, as well as directly compared the High TF and Low TF conditions. Second, I conducted a set of equivalent mixed ANOVAs to assess the robustness of the results.

The results were largely consistent with those of the main analyses, with one exception. In the GLMMs for the testable sample, ratings in the High Type-Frequency condition were neither significantly greater than those in the Low Type-Frequency condition nor than those in the Single Occurrence condition. It is unclear why these results diverge from those based on the difference scores, CLMMs, and ANOVAs.

3 Appendix

4 Descriptives

4.1 Raw ratings

Table 5: Descriptives for the raw ratings. The p value reflects a Wilcoxon test against the scale midpoint of 2.5. Lines in bold face and with a clear background reflect ratings that are significantly above 2.5 in the online sample. Lines in bold face, red font color and a gray background reflect ratings that are significantly below 2.5 in the online sample. Significance codes: *** $p \leq .001$, ** $p \leq .01$, * $p \leq .05$, . $p \leq .10$, ns = not significant.

	Students						Testable					
	N	M	SE	Cohen's d	p		N	M	SE	Cohen's d	p	
First Party Acceptability - bad												
High Type-Frequency	17	1.61	0.094	4.30	0.000	***	41	1.35	0.054	3.98	0.000	***
Low Type-Frequency	17	1.57	0.099	3.98	0.000	***	41	1.32	0.060	3.51	0.000	***
Single Occurrence	17	1.54	0.097	4.00	0.000	***	41	1.30	0.057	3.60	0.000	***
First Party Acceptability - neutral												
High Type-Frequency	17	2.81	0.106	6.64	0.004	**	39	3.10	0.071	7.08	0.000	***
Low Type-Frequency	17	2.59	0.105	6.19	0.705	ns	39	2.97	0.077	6.24	0.000	***
Single Occurrence	17	2.47	0.124	4.96	0.571	ns	39	2.90	0.082	5.70	0.000	***
First Party Acceptability - good												
High Type-Frequency	18	3.69	0.070	12.86	0.000	***	39	3.79	0.043	14.47	0.000	***
Low Type-Frequency	18	3.68	0.075	11.85	0.000	***	39	3.72	0.051	11.83	0.000	***
Single Occurrence	18	3.63	0.083	10.57	0.000	***	39	3.68	0.053	11.33	0.000	***
Third Party Acceptability - bad												
High Type-Frequency	17	2.71	0.146	4.63	0.192	ns	41	2.95	0.129	3.63	0.002	**
Low Type-Frequency	17	1.89	0.121	3.88	0.001	**	41	2.46	0.133	2.92	0.619	ns
Single Occurrence	17	1.66	0.115	3.59	0.000	***	41	1.86	0.091	3.22	0.000	***
Third Party Acceptability - neutral												
High Type-Frequency	17	3.46	0.095	9.14	0.000	***	39	3.58	0.077	7.53	0.000	***
Low Type-Frequency	17	2.72	0.098	6.92	0.038	*	39	3.11	0.095	5.32	0.000	***
Single Occurrence	17	2.32	0.119	4.89	0.097	.	39	2.77	0.088	5.08	0.005	**
Third Party Acceptability - good												
High Type-Frequency	18	3.80	0.060	15.42	0.000	***	39	3.84	0.050	12.52	0.000	***
Low Type-Frequency	18	3.45	0.117	7.16	0.000	***	39	3.65	0.060	9.87	0.000	***
Single Occurrence	18	3.00	0.155	4.70	0.007	**	39	3.34	0.077	7.05	0.000	***
Prevalence - bad												
High Type-Frequency	17	3.53	0.104	8.45	0.000	***	41	3.48	0.098	5.59	0.000	***
Low Type-Frequency	17	2.75	0.164	4.18	0.181	ns	41	2.96	0.118	3.98	0.001	***
Single Occurrence	17	2.09	0.115	4.53	0.005	**	41	2.27	0.070	5.10	0.005	**
Prevalence - neutral												
High Type-Frequency	17	3.51	0.117	7.48	0.000	***	39	3.66	0.078	7.64	0.000	***
Low Type-Frequency	17	2.50	0.091	6.88	0.875	ns	39	2.90	0.106	4.46	0.001	**
Single Occurrence	17	1.96	0.114	4.29	0.001	***	39	2.33	0.095	3.97	0.063	.
Prevalence - good												
High Type-Frequency	18	3.75	0.071	12.91	0.000	***	39	3.79	0.053	11.66	0.000	***

Table 5: Descriptives for the raw ratings. The p value reflects a Wilcoxon test against the scale midpoint of 2.5. Lines in bold face and with a clear background reflect ratings that are significantly above 2.5 in the online sample. Lines in bold face, red font color and a gray background reflect ratings that are significantly below 2.5 in the online sample. Significance codes: *** $p \leq .001$, ** $p \leq .01$, * $p \leq .05$, . $p \leq .10$, ns = not significant. (continued)

	Students						Testable					
	N	M	SE	Cohen's d	p		N	M	SE	Cohen's d	p	
Low Type-Frequency	18	3.29	0.134	5.97	0.000	***	39	3.43	0.073	7.58	0.000	***
Single Occurrence	18	2.70	0.149	4.39	0.303	ns	39	2.96	0.099	4.86	0.000	***

For bad actions, *prevalence ratings* were above the midpoint of 2.5 for low and high type-frequency actions, and below the midpoint for single occurrence actions (though these differences did not reach significance for the student population). For neutral actions, the results were similar, except that single occurrence actions were only marginally below the midpoint in the online sample. In contrast, for good actions, prevalence ratings were above the midpoint in all frequency conditions.

For bad actions, *first-party acceptability* ratings were below the midpoint in all frequency conditions. In contrast, for neutral and good actions, the ratings were above the midpoint (though these effects did not reach significance in the student sample.)

For bad actions, *first-party acceptability* were above the midpoint for high type-frequency actions, below the midpoint for single occurrence actions and did not differ from the midpoint for low type-frequency actions. For good actions, ratings were above the midpoint for all frequency conditions.

Table 6: Results on a set of cumulative linear mixed model, fitted separately for each sample (online or students) and domain (first-party acceptability and third-party acceptability), with the fixed factor predictors valence and frequency condition as well as their interaction, as well as a random slope for participants. Only the main effect of valence is shown. Significance codes: *** $p \leq .001$, ** $p \leq .01$, * $p \leq .05$, . $p \leq .10$, ns = not significant.

Parameter	Coefficient	SE	CI_{low}	CI_{high}	z	p
First Party Acceptability - students						
Valence:bad	-2.72	0.444	-3.589	-1.85	-6.12	0 ***
Valence:good	2.45	0.452	1.561	3.33	5.41	0 ***
Third Party Acceptability - students						
Valence:bad	-2.14	0.428	-2.981	-1.30	-5.00	0 ***
Valence:good	1.62	0.448	0.738	2.49	3.61	0 ***
First Party Acceptability - testable						
Valence:bad	-4.28	0.330	-4.922	-3.63	-12.96	0 ***
Valence:good	2.35	0.339	1.683	3.01	6.92	0 ***
Third Party Acceptability - testable						
Valence:bad	-1.93	0.351	-2.616	-1.24	-5.50	0 ***
Valence:good	1.36	0.376	0.628	2.10	3.63	0 ***

As shown in Table 6, acceptability ratings were higher for good actions than for neutral actions, and for neutral actions than for bad actions.

4.2 Comparisons of the high and low type-frequency conditions against the single occurrence condition

Table 7: Descriptives for the difference scores against the single occurrence condition ($\frac{\text{High Type-Frequency}-\text{Single Occurrence}}{\text{High Type-Frequency}+\text{Single Occurrence}}$ and $\frac{\text{Low Type-Frequency}-\text{Single Occurrence}}{\text{Low Type-Frequency}+\text{Single Occurrence}}$). The p-value reflects a Wilcoxon test against zero. p_{hb} reflects the Holm-Bonferroni corrected p-value. The correction was applied separately for each sample. The likelihood ratios $LR_{01,BIC}$ and $LR_{01,AIC}$ reflect likelihood ratios in favor of the null hypothesis after correction with the Bayesian Information Criterion and the Akaike Information Criterion, respectively. Significance codes: *** $p \leq .001$, ** $p \leq .01$, * $p \leq .05$, . $p \leq .10$, ns = not significant.

Valence	Frequency Condition	N	<i>M</i>	<i>SE</i>	Cohen's <i>d</i>	<i>p</i>	p_{hb}		lr01.bic	lr01.aic
First Party Acceptability - students										
bad	High Type-Frequency	17	0.022	0.010	0.551	0.054	0.178	ns	0.313	0.278
bad	Low Type-Frequency	17	0.007	0.011	0.147	0.551	0.551	ns	3.433	3.040
neutral	High Type-Frequency	17	0.069	0.022	0.776	0.003	0.017	*		
neutral	Low Type-Frequency	17	0.029	0.014	0.530	0.045	0.178	ns	0.377	0.334
good	High Type-Frequency	18	0.010	0.004	0.659	0.014	0.072	.	0.085	0.072
good	Low Type-Frequency	18	0.007	0.005	0.330	0.213	0.426	ns	1.595	1.345
First Party Acceptability - testable										
bad	High Type-Frequency	41	0.022	0.009	0.382	0.024	0.096	.	0.322	0.152
bad	Low Type-Frequency	41	0.009	0.007	0.211	0.165	0.495	ns	2.562	1.210
neutral	High Type-Frequency	39	0.036	0.009	0.622	0.001	0.005	**		
neutral	Low Type-Frequency	39	0.014	0.008	0.281	0.173	0.495	ns	1.342	0.654
good	High Type-Frequency	39	0.016	0.005	0.475	0.004	0.021	*		
good	Low Type-Frequency	39	0.006	0.005	0.176	0.313	0.495	ns	3.402	1.658
Third Party Acceptability - students										
bad	High Type-Frequency	17	0.238	0.046	1.286	0.001	0.009	**		
bad	Low Type-Frequency	17	0.067	0.021	0.802	0.002	0.016	*		
neutral	High Type-Frequency	17	0.201	0.029	1.713	0.000	0.008	**		
neutral	Low Type-Frequency	17	0.084	0.021	0.997	0.002	0.017	*		
good	High Type-Frequency	18	0.126	0.026	1.171	0.000	0.008	**		
good	Low Type-Frequency	18	0.076	0.021	0.889	0.000	0.008	**		
Third Party Acceptability - testable										
bad	High Type-Frequency	41	0.221	0.025	1.418	0.000	0.000	***		
bad	Low Type-Frequency	41	0.129	0.020	1.014	0.000	0.000	***		
neutral	High Type-Frequency	39	0.131	0.020	1.066	0.000	0.000	***		
neutral	Low Type-Frequency	39	0.058	0.019	0.485	0.002	0.010	*		
good	High Type-Frequency	39	0.072	0.012	0.970	0.000	0.000	***		
good	Low Type-Frequency	39	0.047	0.010	0.794	0.000	0.000	***		
Prevalence - students										
bad	High Type-Frequency	17	0.259	0.035	1.856	0.000	0.008	**		
bad	Low Type-Frequency	17	0.133	0.030	1.120	0.001	0.009	**		
neutral	High Type-Frequency	17	0.284	0.040	1.773	0.000	0.008	**		
neutral	Low Type-Frequency	17	0.130	0.025	1.299	0.001	0.009	**		
good	High Type-Frequency	18	0.172	0.026	1.580	0.000	0.004	**		
good	Low Type-Frequency	18	0.102	0.024	1.031	0.001	0.010	**		
Prevalence - testable										

Table 7: Descriptives for the difference scores against the single occurrence condition ($\frac{\text{High Type-Frequency}-\text{Single Occurrence}}{\text{High Type-Frequency}+\text{Single Occurrence}}$ and $\frac{\text{Low Type-Frequency}-\text{Single Occurrence}}{\text{Low Type-Frequency}+\text{Single Occurrence}}$). The p-value reflects a Wilcoxon test against zero. p_{hb} reflects the Holm-Bonferroni corrected p-value. The correction was applied separately for each sample. The likelihood ratios $LR_{01,BIC}$ and $LR_{01,AIC}$ reflect likelihood ratios in favor of the null hypothesis after correction with the Bayesian Information Criterion and the Akaike Information Criterion, respectively. Significance codes: *** $p \leq .001$, ** $p \leq .01$, * $p \leq .05$, . $p \leq .10$, ns = not significant. (*continued*)

Valence	Frequency Condition	N	M	SE	Cohen's d	p	p_{hb}		lr01.bic	lr01.aic
bad	High Type-Frequency	41	0.207	0.018	1.803	0.000	0.000	***		
bad	Low Type-Frequency	41	0.123	0.017	1.151	0.000	0.000	***		
neutral	High Type-Frequency	39	0.227	0.024	1.521	0.000	0.000	***		
neutral	Low Type-Frequency	39	0.111	0.018	0.977	0.000	0.000	***		
good	High Type-Frequency	39	0.131	0.018	1.176	0.000	0.000	***		
good	Low Type-Frequency	39	0.079	0.015	0.851	0.000	0.000	***		

4.3 Comparisons of the high and low type-frequency conditions

Table 8: Descriptive statistics for the difference scores comparing the High and Low Type-Frequency Conditions ($\frac{\text{High Type-Frequency} - \text{Low Type-Frequency}}{\text{High Type-Frequency} + \text{Low Type-Frequency}}$). The p-value reflects a Wilcoxon test against zero. p_{hb} reflects the Holm-Bonferroni corrected p-value. The correction was applied separately for each sample. Significance codes: *** $p \leq .001$, ** $p \leq .01$, * $p \leq .05$, . $p \leq .10$, ns = not significant.

Valence	N	<i>M</i>	<i>SE</i>	Cohen's <i>d</i>	<i>p</i>	<i>p</i> _{hb}	
First Party Acceptability - Students							
bad	17	0.016	0.013	0.307	0.476	0.740	ns
neutral	17	0.041	0.014	0.736	0.010	0.031	*
good	18	0.003	0.003	0.227	0.370	0.740	ns
First Party Acceptability - Testable							
bad	41	0.013	0.009	0.236	0.133	0.133	ns
neutral	39	0.022	0.006	0.588	0.000	0.001	**
good	39	0.010	0.003	0.539	0.002	0.005	**
Third Party Acceptability - Students							
bad	17	0.176	0.042	1.038	0.001	0.005	**
neutral	17	0.121	0.020	1.488	0.000	0.003	**
good	18	0.051	0.017	0.728	0.008	0.031	*
Third Party Acceptability - Testable							
bad	41	0.096	0.021	0.718	0.000	0.000	***
neutral	39	0.074	0.014	0.838	0.000	0.000	***
good	39	0.026	0.007	0.571	0.001	0.003	**
Prevalence - Students							
bad	17	0.132	0.035	0.947	0.004	0.019	*
neutral	17	0.165	0.028	1.477	0.000	0.004	**
good	18	0.071	0.020	0.863	0.001	0.005	**
Prevalence - Testable							
bad	41	0.086	0.017	0.803	0.000	0.000	***
neutral	39	0.122	0.019	1.012	0.000	0.000	***
good	39	0.053	0.012	0.735	0.000	0.000	***

5 Alternative analyses

5.1 Comparisons of the high and low type-frequency conditions against the single occurrence condition

5.1.1 GLMMs

I analyzed the results using a series of generalized linear mixed models (GLMMs), in which responses were converted to binary outcomes on each trial: ratings of 1 or 2 were coded as “unacceptable,” and ratings of 3 or 4 as “acceptable.” The models included fixed factor predictors for Frequency Condition and Valence, their interaction, and a random slope for participants. The results are shown in Table 9.

Table 9: Results from a set of generalized linear mixed models, fitted to the binarized trial-by-trial data, separately for each sample (online or student), domain (First-Party Acceptability, Third-Party Acceptability, and Prevalence), and frequency contrast (High Type-Frequency vs. Single Occurrence and Low Type-Frequency vs. Single Occurrence). The models included the fixed factor predictors Valence and Frequency Condition, their interaction, and random slopes for participants and actions. Significance codes: *** $p \leq .001$, ** $p \leq .01$, * $p \leq .05$, . $p \leq .10$, ns = not significant.

Parameter	Coefficient	<i>SE</i>	<i>CI</i>	<i>CI</i> _{low}	<i>CI</i> _{high}	<i>z</i>	<i>p</i>
First Party Acceptability - High Type-Frequency (vs. Single Occurrence) - Students							
(Intercept)	0.092	0.682	0.95	-1.245	1.428	0.134	0.893 ns
Valence:bad	-4.109	1.050	0.95	-6.166	-2.052	-3.915	0.000 ***
Valence:good	4.535	1.088	0.95	2.403	6.666	4.169	0.000 ***
Freq. Cond:high Type-Freq.	0.756	0.273	0.95	0.220	1.292	2.766	0.006 **
Valence:bad×Freq. Cond:high Type-Freq.	-0.160	0.499	0.95	-1.138	0.818	-0.321	0.749 ns
Valence:good×Freq. Cond:high Type-Freq.	-0.539	0.540	0.95	-1.598	0.520	-0.998	0.318 ns
First Party Acceptability - High Type-Frequency (vs. Single Occurrence) - Testable							
(Intercept)	1.363	0.509	0.95	0.365	2.362	2.676	0.007 **
Valence:bad	-5.529	0.776	0.95	-7.050	-4.008	-7.126	0.000 ***
Valence:good	3.247	0.790	0.95	1.699	4.794	4.112	0.000 ***
Freq. Cond:high Type-Freq.	0.228	0.188	0.95	-0.140	0.595	1.215	0.224 ns
Valence:bad×Freq. Cond:high Type-Freq.	-0.330	0.370	0.95	-1.055	0.396	-0.890	0.373 ns
Valence:good×Freq. Cond:high Type-Freq.	0.177	0.415	0.95	-0.636	0.990	0.426	0.670 ns
First Party Acceptability - Low Type-Frequency (vs. Single Occurrence) - Students							
(Intercept)	0.113	0.699	0.95	-1.257	1.483	0.162	0.872 ns
Valence:bad	-3.960	1.061	0.95	-6.039	-1.881	-3.734	0.000 ***
Valence:good	4.618	1.116	0.95	2.430	6.806	4.137	0.000 ***
Freq. Cond:low Type-Freq.	0.311	0.280	0.95	-0.237	0.858	1.111	0.267 ns
Valence:bad×Freq. Cond:low Type-Freq.	-0.133	0.506	0.95	-1.124	0.858	-0.263	0.792 ns
Valence:good×Freq. Cond:low Type-Freq.	0.028	0.553	0.95	-1.055	1.112	0.051	0.959 ns
First Party Acceptability - Low Type-Frequency (vs. Single Occurrence) - Testable							
(Intercept)	1.469	0.577	0.95	0.338	2.600	2.546	0.011 *
Valence:bad	-6.318	0.903	0.95	-8.087	-4.548	-6.998	0.000 ***
Valence:good	3.489	0.891	0.95	1.743	5.235	3.917	0.000 ***
Freq. Cond:low Type-Freq.	-0.019	0.193	0.95	-0.397	0.360	-0.097	0.923 ns
Valence:bad×Freq. Cond:low Type-Freq.	-0.043	0.400	0.95	-0.827	0.741	-0.107	0.915 ns
Valence:good×Freq. Cond:low Type-Freq.	0.019	0.403	0.95	-0.772	0.809	0.046	0.963 ns
Third Party Acceptability - High Type-Frequency (vs. Single Occurrence) - Students							
(Intercept)	-0.545	0.335	0.95	-1.202	0.112	-1.627	0.104 ns

Table 9: Results from a set of generalized linear mixed models, fitted to the binarized trial-by-trial data, separately for each sample (online or student), domain (First-Party Acceptability, Third-Party Acceptability, and Prevalence), and frequency contrast (High Type-Frequency vs. Single Occurrence and Low Type-Frequency vs. Single Occurrence). The models included the fixed factor predictors Valence and Frequency Condition, their interaction, and random slopes for participants and actions. Significance codes: *** $p \leq .001$, ** $p \leq .01$, * $p \leq .05$, . $p \leq .10$, ns = not significant. (continued)

Parameter	Coefficient	SE	CI	CI _{low}	CI _{high}	z	p	
Valence:bad	-2.063	0.516	0.95	-3.075	-1.050	-3.994	0.000	***
Valence:good	1.564	0.481	0.95	0.621	2.507	3.250	0.001	**
Freq. Cond:high Type-Freq.	2.855	0.289	0.95	2.288	3.422	9.865	0.000	***
Valence:bad×Freq. Cond:high Type-Freq.	0.449	0.423	0.95	-0.380	1.279	1.062	0.288	ns
Valence:good×Freq. Cond:high Type-Freq.	0.676	0.585	0.95	-0.470	1.822	1.156	0.248	ns
Third Party Acceptability - High Type-Frequency (vs. Single Occurrence) - Testable								
(Intercept)	0.842	0.345	0.95	0.165	1.519	2.437	0.015	*
Valence:bad	-2.755	0.493	0.95	-3.722	-1.788	-5.585	0.000	***
Valence:good	1.858	0.516	0.95	0.848	2.868	3.604	0.000	***
Freq. Cond:high Type-Freq.	2.169	0.219	0.95	1.739	2.598	9.901	0.000	***
Valence:bad×Freq. Cond:high Type-Freq.	1.158	0.302	0.95	0.566	1.749	3.835	0.000	***
Valence:good×Freq. Cond:high Type-Freq.	0.225	0.435	0.95	-0.627	1.078	0.519	0.604	ns
Third Party Acceptability - Low Type-Frequency (vs. Single Occurrence) - Students								
(Intercept)	-0.581	0.389	0.95	-1.343	0.182	-1.492	0.136	ns
Valence:bad	-2.217	0.592	0.95	-3.378	-1.057	-3.746	0.000	***
Valence:good	1.724	0.556	0.95	0.634	2.814	3.100	0.002	**
Freq. Cond:low Type-Freq.	0.959	0.229	0.95	0.510	1.408	4.187	0.000	***
Valence:bad×Freq. Cond:low Type-Freq.	0.085	0.386	0.95	-0.671	0.840	0.220	0.826	ns
Valence:good×Freq. Cond:low Type-Freq.	0.895	0.382	0.95	0.145	1.644	2.339	0.019	*
Third Party Acceptability - Low Type-Frequency (vs. Single Occurrence) - Testable								
(Intercept)	0.887	0.397	0.95	0.109	1.664	2.236	0.025	*
Valence:bad	-3.006	0.569	0.95	-4.120	-1.891	-5.287	0.000	***
Valence:good	1.937	0.587	0.95	0.785	3.088	3.297	0.001	***
Freq. Cond:low Type-Freq.	0.914	0.186	0.95	0.550	1.278	4.918	0.000	***
Valence:bad×Freq. Cond:low Type-Freq.	0.952	0.269	0.95	0.424	1.479	3.538	0.000	***
Valence:good×Freq. Cond:low Type-Freq.	0.744	0.353	0.95	0.052	1.436	2.106	0.035	*
Prevalence - High Type-Frequency (vs. Single Occurrence) - Students								
(Intercept)	-1.494	0.332	0.95	-2.144	-0.843	-4.502	0.000	***
Valence:bad	0.044	0.472	0.95	-0.881	0.968	0.092	0.927	ns
Valence:good	1.792	0.460	0.95	0.890	2.694	3.894	0.000	***
Freq. Cond:high Type-Freq.	3.696	0.307	0.95	3.094	4.298	12.034	0.000	***
Valence:bad×Freq. Cond:high Type-Freq.	0.872	0.485	0.95	-0.079	1.823	1.797	0.072	.
Valence:good×Freq. Cond:high Type-Freq.	0.504	0.589	0.95	-0.651	1.659	0.856	0.392	ns
Prevalence - High Type-Frequency (vs. Single Occurrence) - Testable								
(Intercept)	-0.736	0.339	0.95	-1.400	-0.072	-2.172	0.030	*
Valence:bad	-0.317	0.475	0.95	-1.248	0.615	-0.666	0.505	ns
Valence:good	2.063	0.489	0.95	1.104	3.021	4.217	0.000	***
Freq. Cond:high Type-Freq.	4.163	0.266	0.95	3.641	4.685	15.628	0.000	***
Valence:bad×Freq. Cond:high Type-Freq.	-0.101	0.358	0.95	-0.803	0.600	-0.283	0.777	ns
Valence:good×Freq. Cond:high Type-Freq.	-0.803	0.408	0.95	-1.603	-0.004	-1.970	0.049	*

Table 9: Results from a set of generalized linear mixed models, fitted to the binarized trial-by-trial data, separately for each sample (online or student), domain (First-Party Acceptability, Third-Party Acceptability, and Prevalence), and frequency contrast (High Type-Frequency vs. Single Occurrence and Low Type-Frequency vs. Single Occurrence). The models included the fixed factor predictors Valence and Frequency Condition, their interaction, and random slopes for participants and actions. Significance codes: *** $p \leq .001$, ** $p \leq .01$, * $p \leq .05$, . $p \leq .10$, ns = not significant. (continued)

Parameter	Coefficient	SE	CI	CI _{low}	CI _{high}	z	p
Prevalence - Low Type-Frequency (vs. Single Occurrence) - Students							
(Intercept)	-1.777	0.507	0.95	-2.771	-0.783	-3.505	0.000 ***
Valence:bad	-0.329	0.729	0.95	-1.758	1.099	-0.452	0.652 ns
Valence:good	2.194	0.707	0.95	0.809	3.579	3.105	0.002 **
Freq. Cond:low Type-Freq.	1.337	0.251	0.95	0.846	1.829	5.331	0.000 ***
Valence:bad×Freq. Cond:low Type-Freq.	0.859	0.391	0.95	0.092	1.626	2.195	0.028 *
Valence:good×Freq. Cond:low Type-Freq.	0.663	0.380	0.95	-0.083	1.408	1.742	0.081 .
Prevalence - Low Type-Frequency (vs. Single Occurrence) - Testable							
(Intercept)	-0.998	0.406	0.95	-1.795	-0.202	-2.457	0.014 *
Valence:bad	-0.221	0.568	0.95	-1.334	0.892	-0.389	0.697 ns
Valence:good	2.492	0.580	0.95	1.355	3.630	4.294	0.000 ***
Freq. Cond:low Type-Freq.	1.935	0.196	0.95	1.550	2.320	9.849	0.000 ***
Valence:bad×Freq. Cond:low Type-Freq.	0.335	0.272	0.95	-0.199	0.868	1.229	0.219 ns
Valence:good×Freq. Cond:low Type-Freq.	0.498	0.311	0.95	-0.111	1.107	1.602	0.109 ns

When comparing the High Type-Frequency and Low Type-Frequency conditions to the Single Occurrence condition, the results of the GLMMs were similar to those of the CLMMs. However, for first-party acceptability in the testable sample, the effect of Frequency Condition was not significant in the High Type-Frequency condition. In the student sample, and as in the CLMMs, ratings differed from the Single Occurrence condition only in the High Type-Frequency condition, and not in the Low Type-Frequency condition.

For third-party acceptability and prevalence, both the High and Low Type-Frequency conditions differed significantly from the Single Occurrence condition, although parameter estimates were approximately two to three times larger in the High Type-Frequency condition.

I also observed main effects of valence, indicating that good actions were rated higher than neutral ones, and neutral actions higher than bad ones—except in prevalence ratings, where neutral and bad actions did not differ.

5.1.2 ANOVAs

The results were also analyzed using a series of mixed ANOVAs, with Frequency Condition (High Type-Frequency vs. Single Occurrence and Low Type-Frequency vs. Single Occurrence) as a within-participant factor, and Valence as a between-participant factor, including their interaction.

Table 10: Results from a set of mixed ANOVAs, fitted separately for each sample (online or student), domain (First-Party Acceptability, Third-Party Acceptability, and Prevalence), and frequency contrast (High Type-Frequency vs. Single Occurrence and Low Type-Frequency vs. Single Occurrence). The models included Valence (between-subjects) and Frequency Condition (within-subjects) as fixed factors, along with their interaction. Significance codes: *** $p \leq .001$, ** $p \leq .01$, * $p \leq .05$, . $p \leq .10$, ns = not significant.

Effect	df	MSE	F	ges	p	
First Party Acceptability - High Type-Frequency (vs. Single Occurrence) - Students						
Valence	2, 49	0.26	144.33 ***	.836	<.001	***
Freq. Cond	1, 49	0.04	16.63 ***	.043	<.001	***
Valence×Freq. Cond	2, 49	0.04	5.63 **	.030	.006	***
First Party Acceptability - High Type-Frequency (vs. Single Occurrence) - Testable						
Valence	2, 116	0.26	472.95 ***	.880	<.001	***
Freq. Cond	1, 116	0.03	30.27 ***	.026	<.001	***
Valence×Freq. Cond	2, 116	0.03	3.55 *	.006	.032	***
First Party Acceptability - Low Type-Frequency (vs. Single Occurrence) - Students						
Valence	2, 49	0.30	130.42 ***	.835	<.001	***
Freq. Cond	1, 49	0.02	7.08 *	.007	.011	***
Valence×Freq. Cond	2, 49	0.02	1.57	.003	.219	***
First Party Acceptability - Low Type-Frequency (vs. Single Occurrence) - Testable						
Valence	2, 116	0.30	402.14 ***	.866	<.001	***
Freq. Cond	1, 116	0.02	5.86 *	.003	.017	***
Valence×Freq. Cond	2, 116	0.02	0.51	<.001	.604	***
Third Party Acceptability - High Type-Frequency (vs. Single Occurrence) - Students						
Valence	2, 49	0.21	62.40 ***	.533	<.001	***
Freq. Cond	1, 49	0.26	101.14 ***	.532	<.001	***
Valence×Freq. Cond	2, 49	0.26	1.07	.024	.350	***
Third Party Acceptability - High Type-Frequency (vs. Single Occurrence) - Testable						
Valence	2, 116	0.40	73.16 ***	.447	<.001	***
Freq. Cond	1, 116	0.22	171.69 ***	.347	<.001	***
Valence×Freq. Cond	2, 116	0.22	7.92 ***	.047	<.001	***
Third Party Acceptability - Low Type-Frequency (vs. Single Occurrence) - Students						
Valence	2, 49	0.41	45.12 ***	.605	<.001	***
Freq. Cond	1, 49	0.08	41.00 ***	.125	<.001	***
Valence×Freq. Cond	2, 49	0.08	1.46	.010	.243	***
Third Party Acceptability - Low Type-Frequency (vs. Single Occurrence) - Testable						
Valence	2, 116	0.53	68.55 ***	.476	<.001	***
Freq. Cond	1, 116	0.16	64.89 ***	.115	<.001	***
Valence×Freq. Cond	2, 116	0.16	3.18 *	.013	.045	***
Prevalence - High Type-Frequency (vs. Single Occurrence) - Students						
Valence	2, 49	0.15	16.46 ***	.192	<.001	***
Freq. Cond	1, 49	0.28	171.08 ***	.693	<.001	***
Valence×Freq. Cond	2, 49	0.28	2.15	.054	.128	***
Prevalence - High Type-Frequency (vs. Single Occurrence) - Testable						
Valence	2, 116	0.29	18.46 ***	.145	<.001	***
Freq. Cond	1, 116	0.25	295.96 ***	.544	<.001	***
Valence×Freq. Cond	2, 116	0.25	5.01 **	.039	.008	***

Table 10: Results from a set of mixed ANOVAs, fitted separately for each sample (online or student), domain (First-Party Acceptability, Third-Party Acceptability, and Prevalence), and frequency contrast (High Type-Frequency vs. Single Occurrence and Low Type-Frequency vs. Single Occurrence). The models included Valence (between-subjects) and Frequency Condition (within-subjects) as fixed factors, along with their interaction. Significance codes: *** $p \leq .001$, ** $p \leq .01$, * $p \leq .05$, . $p \leq .10$, ns = not significant. (*continued*)

Effect	df	MSE	F	ges	p	
Prevalence - Low Type-Frequency (vs. Single Occurrence) - Students						
Valence	2, 49	0.41	13.61 ***	.288	<.001	***
Freq. Cond	1, 49	0.15	61.30 ***	.253	<.001	***
Valence×Freq. Cond	2, 49	0.15	0.18	.002	.834	***
Prevalence - Low Type-Frequency (vs. Single Occurrence) - Testable						
Valence	2, 116	0.53	16.39 ***	.177	<.001	***
Freq. Cond	1, 116	0.17	117.88 ***	.196	<.001	***
Valence×Freq. Cond	2, 116	0.17	1.38	.006	.256	***

As shown in 10, the results of these ANOVAs were similar to those from the CLMMs. As in the CLMMs, there was a main effect of Frequency Condition. However, the effect was also significant for the Low Type-Frequency condition, though the effect size was approximately six to eight times greater for the High Type-Frequency condition.

There was also a main effect of Valence, as well as a significant interaction between Valence and Frequency Condition across all domains and frequency contrasts.

5.2 Comparisons of the high and low type-frequency conditions

5.2.1 GLMMs

I also analyzed the results using a series of generalized linear mixed models (GLMMs), in which responses were converted to binary outcomes on each trial: ratings of 1 or 2 were coded as “unacceptable,” and ratings of 3 or 4 as “acceptable.” The models included fixed factor predictors for Frequency Condition and Valence, their interaction, and a random slope for participants and actions.

When comparing the High Type-Frequency condition to the Low Type-Frequency condition, the results of the GLMMs were broadly similar to those obtained from the CLMMs and the mixed ANOVAs, with one exception. For first-party acceptability, the effect of Frequency Condition was not significant in the testable sample and only marginal in the student sample. It is unclear why this analysis diverges from the results of both the CLMMs and the ANOVAs (see below).

For third-party acceptability and prevalence, the effect of Frequency Condition was significant in both samples.

I also observed main effects of Valence, indicating that good actions were rated higher than neutral ones, and neutral actions higher than bad ones, except in prevalence ratings, where neutral and bad actions did not differ.

Table 11: Results from a set of generalized linear mixed models using binarized data (ratings of 1–2 coded as unacceptable, and 3–4 as acceptable), comparing the High Type-Frequency condition to the Low Type-Frequency condition. Models were fitted separately for each sample (online or student) and domain (First-Party Acceptability and Third-Party Acceptability), with fixed factor predictors Valence and Frequency Condition, their interaction, and a random slope for participants. Significance codes: *** $p \leq .001$, ** $p \leq .01$, * $p \leq .05$, . $p \leq .10$, ns = not significant.

Parameter	Coefficient	SE	CI	CI _{low}	CI _{high}	z	p
First Party Acceptability - Students							
(Intercept)	0.455	0.694	0.95	-0.905	1.816	0.656	0.512 ns
Valence:bad	-4.110	1.044	0.95	-6.157	-2.064	-3.937	0.000 ***
Valence:good	4.580	1.123	0.95	2.379	6.781	4.078	0.000 ***
Freq. Cond:high Type-Freq.	0.490	0.277	0.95	-0.053	1.032	1.770	0.077 .
Valence:bad×Freq. Cond:high Type-Freq.	-0.097	0.485	0.95	-1.047	0.853	-0.200	0.841 ns
Valence:good×Freq. Cond:high Type-Freq.	-0.608	0.561	0.95	-1.707	0.490	-1.085	0.278 ns
First Party Acceptability - Testable							
(Intercept)	1.483	0.609	0.95	0.288	2.677	2.433	0.015 *
Valence:bad	-6.194	0.934	0.95	-8.024	-4.363	-6.633	0.000 ***
Valence:good	3.908	0.963	0.95	2.020	5.796	4.057	0.000 ***
Freq. Cond:high Type-Freq.	0.279	0.200	0.95	-0.113	0.671	1.393	0.163 ns
Valence:bad×Freq. Cond:high Type-Freq.	-0.335	0.390	0.95	-1.099	0.429	-0.859	0.390 ns
Valence:good×Freq. Cond:high Type-Freq.	0.166	0.436	0.95	-0.690	1.021	0.379	0.705 ns
Third Party Acceptability - Students							
(Intercept)	0.347	0.318	0.95	-0.275	0.970	1.093	0.274 ns
Valence:bad	-1.983	0.468	0.95	-2.900	-1.065	-4.236	0.000 ***
Valence:good	2.262	0.499	0.95	1.284	3.239	4.535	0.000 ***
Freq. Cond:high Type-Freq.	1.976	0.281	0.95	1.425	2.527	7.028	0.000 ***
Valence:bad×Freq. Cond:high Type-Freq.	0.386	0.384	0.95	-0.367	1.139	1.005	0.315 ns
Valence:good×Freq. Cond:high Type-Freq.	-0.226	0.591	0.95	-1.384	0.932	-0.382	0.702 ns
Third Party Acceptability - Testable							
(Intercept)	2.004	0.457	0.95	1.108	2.901	4.382	0.000 ***
Valence:bad	-2.214	0.627	0.95	-3.443	-0.985	-3.530	0.000 ***
Valence:good	3.213	0.734	0.95	1.775	4.652	4.378	0.000 ***
Freq. Cond:high Type-Freq.	1.546	0.235	0.95	1.085	2.007	6.569	0.000 ***
Valence:bad×Freq. Cond:high Type-Freq.	0.337	0.301	0.95	-0.253	0.927	1.120	0.263 ns
Valence:good×Freq. Cond:high Type-Freq.	-0.629	0.524	0.95	-1.657	0.398	-1.200	0.230 ns
Prevalence - Students							
(Intercept)	-0.310	0.392	0.95	-1.078	0.457	-0.792	0.428 ns
Valence:bad	0.612	0.566	0.95	-0.496	1.721	1.083	0.279 ns
Valence:good	2.512	0.596	0.95	1.344	3.680	4.216	0.000 ***
Freq. Cond:high Type-Freq.	2.631	0.288	0.95	2.067	3.194	9.150	0.000 ***
Valence:bad×Freq. Cond:high Type-Freq.	0.694	0.473	0.95	-0.233	1.621	1.467	0.142 ns
Valence:good×Freq. Cond:high Type-Freq.	0.147	0.618	0.95	-1.063	1.358	0.239	0.811 ns
Prevalence - Testable							
(Intercept)	1.200	0.418	0.95	0.380	2.020	2.868	0.004 **
Valence:bad	-0.005	0.573	0.95	-1.127	1.118	-0.008	0.993 ns
Valence:good	2.965	0.642	0.95	1.707	4.224	4.618	0.000 ***
Freq. Cond:high Type-Freq.	2.768	0.248	0.95	2.282	3.253	11.176	0.000 ***
Valence:bad×Freq. Cond:high Type-Freq.	-0.539	0.333	0.95	-1.192	0.114	-1.617	0.106 ns

Table 11: Results from a set of generalized linear mixed models using binarized data (ratings of 1–2 coded as unacceptable, and 3–4 as acceptable), comparing the High Type-Frequency condition to the Low Type-Frequency condition. Models were fitted separately for each sample (online or student) and domain (First-Party Acceptability and Third-Party Acceptability), with fixed factor predictors Valence and Frequency Condition, their interaction, and a random slope for participants. Significance codes: *** $p \leq .001$, ** $p \leq .01$, * $p \leq .05$, . $p \leq .10$, ns = not significant. (*continued*)

Parameter	Coefficient	SE	CI	CI _{low}	CI _{high}	z	p	
Valence:good×Freq. Cond:high Type-Freq.	-1.630	0.426	0.95	-2.466	-0.795	-3.823	0.000	***

5.2.2 ANOVAs

Table 12: Results from a set of mixed ANOVAs comparing the High Type-Frequency condition to the Low Type-Frequency condition, fitted separately for each sample (online or student) and domain (First-Party Acceptability and Third-Party Acceptability). The models included Valence (between-subjects) and Frequency Condition (within-subjects) as fixed factors, along with their interaction. Significance codes: *** $p \leq .001$, ** $p \leq .01$, * $p \leq .05$, . $p \leq .10$, ns = not significant.

Effect	df	MSE	F	ges	p	
First Party Acceptability - Students						
Valence	2, 49	0.25	151.90 ***	.851	<.001	***
Freq. Cond	1, 49	0.02	10.57 **	.017	.002	***
Valence×Freq. Cond	2, 49	0.02	4.76 *	.015	.013	***
First Party Acceptability - Testable						
Valence	2, 116	0.27	463.66 ***	.884	<.001	***
Freq. Cond	1, 116	0.01	26.05 ***	.010	<.001	***
Valence×Freq. Cond	2, 116	0.01	3.64 *	.003	.029	***
Third Party Acceptability - Students						
Valence	2, 49	0.21	75.86 ***	.621	<.001	***
Freq. Cond	1, 49	0.18	57.22 ***	.355	<.001	***
Valence×Freq. Cond	2, 49	0.18	3.15 +	.057	.052	***
Third Party Acceptability - Testable						
Valence	2, 116	0.59	37.19 ***	.342	<.001	***
Freq. Cond	1, 116	0.14	62.53 ***	.093	<.001	***
Valence×Freq. Cond	2, 116	0.14	4.07 *	.013	.020	***
Prevalence - Students						
Valence	2, 49	0.22	11.59 ***	.186	<.001	***
Freq. Cond	1, 49	0.23	63.62 ***	.401	<.001	***
Valence×Freq. Cond	2, 49	0.23	2.77 +	.055	.073	***
Prevalence - Testable						
Valence	2, 116	0.45	7.77 ***	.086	<.001	***
Freq. Cond	1, 116	0.19	95.12 ***	.194	<.001	***
Valence×Freq. Cond	2, 116	0.19	4.16 *	.021	.018	***

As shown in Table 12, results from a set of mixed ANOVAs comparing the High Type-Frequency condition to the Low Type-Frequency condition, fitted separately for each sample (online or student) and domain (First-Party Acceptability and Third-Party Acceptability), were similar to those of the CLMMs reported above. The models included Valence (between-subjects) and Frequency Condition (within-subjects) as fixed factors, along with their interaction. There were significant main effects of Frequency Condition and Valence across all domains and samples, and the interaction between them was significant in most conditions.

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